Uncertainty and failure analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Recall and choose your starting point

> Record your starting explanation for Uncertainty and failure analysis before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Embeddings and neighbourhoods

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training a small image model

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

A model can be confidently wrong. Use these supplied **output**s for a fictional image classifier: class 1 means a triangle is present; class 0 means it is absent.

The values p1 are probability estimates for class 1, invented for this exercise rather than produced by a trained model. Predict 1 when p1 >= 0.5 and 0 otherwise. The selected-class confidence is max(p1, 1 - p1).

Case and image	p1	Reference label
A clear	0.95	1
B blurred	0.90	0

Case and image	p1	Reference label
C clear	0.80	1
D blurred	0.60	0
E clear	0.45	0
F blurred	0.10	1

A predicts 1 with confidence 0.95 and is correct. F predicts 0 with confidence 0.90 and is wrong: p1 = 0.10 is not low confidence in the selected class.

Reference labels are supplied for this exercise; real investigations must check their quality too.

A score alone cannot establish reliability. **Calibration** asks whether, across many comparable cases, **prediction**s assigned a probability such as 0.90 are correct about 90 percent of the time.

One correct or incorrect case cannot answer that question.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Uncertainty and failure analysis trace and record the first step you can explain without help.

Use W2: Count both types of error for Uncertainty and failure analysis. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Uncertainty and failure analysis explanation against the supplied facts before discussing it.

Use the worked case for Uncertainty and failure analysis. Proposed explanation: “A low p1 always means low confidence”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Analyse new outputs independently

Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Separate confidence from correctness

> Separate confidence from correctness. Apply the supplied Uncertainty and failure analysis facts and keep your reasoning.

For B, E and F, give the predicted class, selected-class confidence and correctness. Explain why p1 = 0.10 does not mean F has low confidence in its predicted class.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Count both types of error

Count both types of error. Apply the supplied Uncertainty and failure analysis facts and keep your reasoning.

For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the review policy

Audit the review policy. Apply the supplied Uncertainty and failure analysis facts and keep your reasoning.

At cutoff 0.80, list accepted and deferred cases, **coverage** and accepted-case accuracy. Can you calculate final accuracy after human review from the supplied data?

What does the report function return for accepted-case accuracy at cutoff 0.99?

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Analyse new outputs independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Analyse new outputs independently. Apply the supplied Uncertainty and failure analysis facts and keep your reasoning.

Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a failure investigation

> Design a failure investigation. Apply the supplied Uncertainty and failure analysis facts and

> keep your reasoning.

All three blurred cases are wrong. A team claims blur caused the errors and plans to put clear images in **training** and their blurred copies in final evaluation.

Identify the unsupported claim and the split problem; propose a better investigation.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Uncertainty and failure analysis to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled uncertainty and failure analysis **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish a predicted class, **confidence score** and observed correctness.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Uncertainty and failure analysis

> Keep positive-class probability separate from confidence in the chosen label. Explain the labelled step before checking the changed condition.

Keep positive-class probability separate from confidence in the chosen label.

Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/29

Positive probability is not always chosen-label confidence

	p(class 1)	prediction	confidence	reference
A	0.9	1	0.9	0
B	0.6	1	0.6	1
C	0.2	0	0.8	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/29

Uncertainty and failure analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predictions

A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.



2. Selected-label confidence

Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite p(class 1)=0.2.



3. Selective review

Accept confidence>=0.8: A and C; accuracy 1/2 and coverage 2/3.

CHANGE ONE THING

Raise the acceptance cutoff to 0.85.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep positive-class probability separate from confidence in the chosen label.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

1. Selected-label confidence

2. Selective review

3. **Prediction**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Selective review Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Raise the acceptance cutoff to 0.85. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Only A is accepted: accuracy 0/1, **coverage** 1/3.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Only A is accepted: accuracy 0/1, **coverage** 1/3. The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Repair a missing building block

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Uncertainty and failure analysis picture from its labels, then record a relevant limit.

01. Uncertainty and failure analysis: the case

AI ACADEMY / AI-5/29 · 01 · LOOK AT THE FACTS

Uncertainty and failure analysis: the case

The situation

Keep positive-class probability separate from confidence in the chosen label.

Supplied evidence

Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p). All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain uncertainty and failure analysis using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

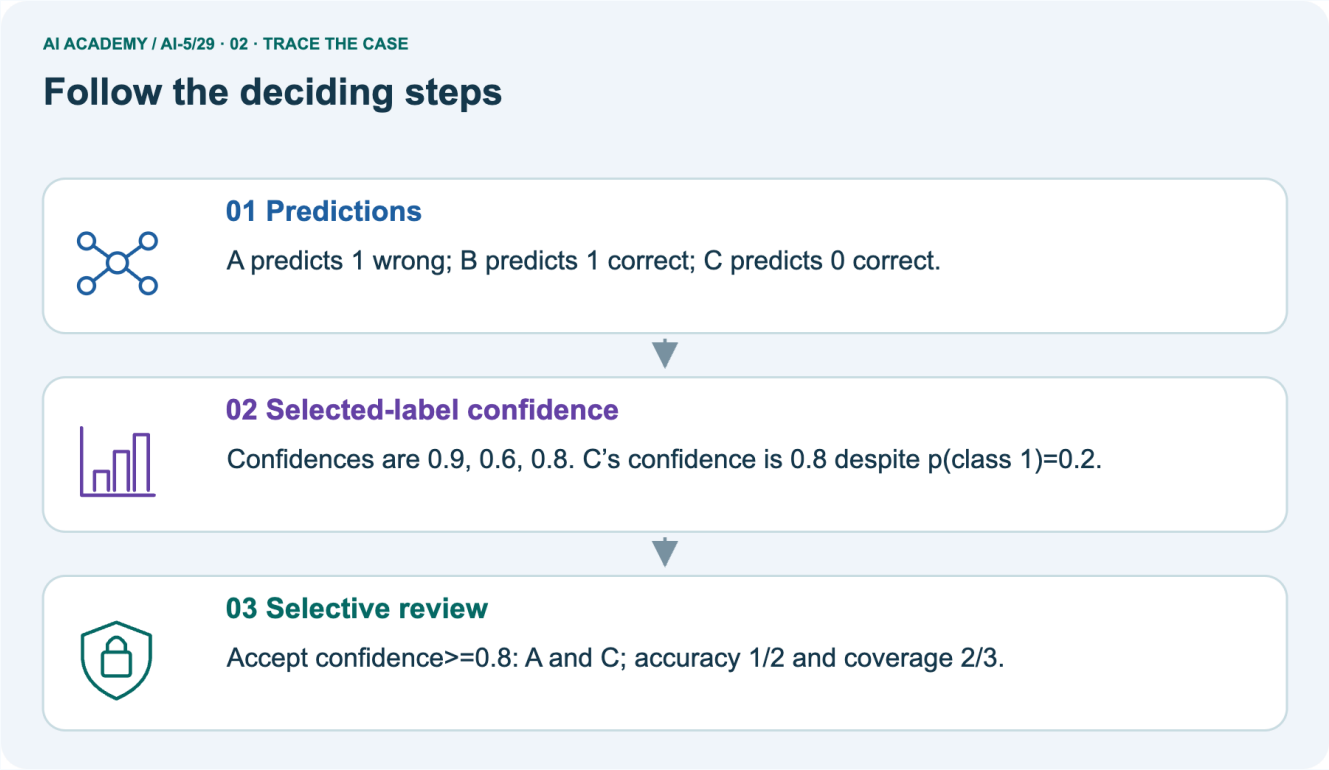
Keep positive-class probability separate from confidence in the chosen label. Supplied facts: Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

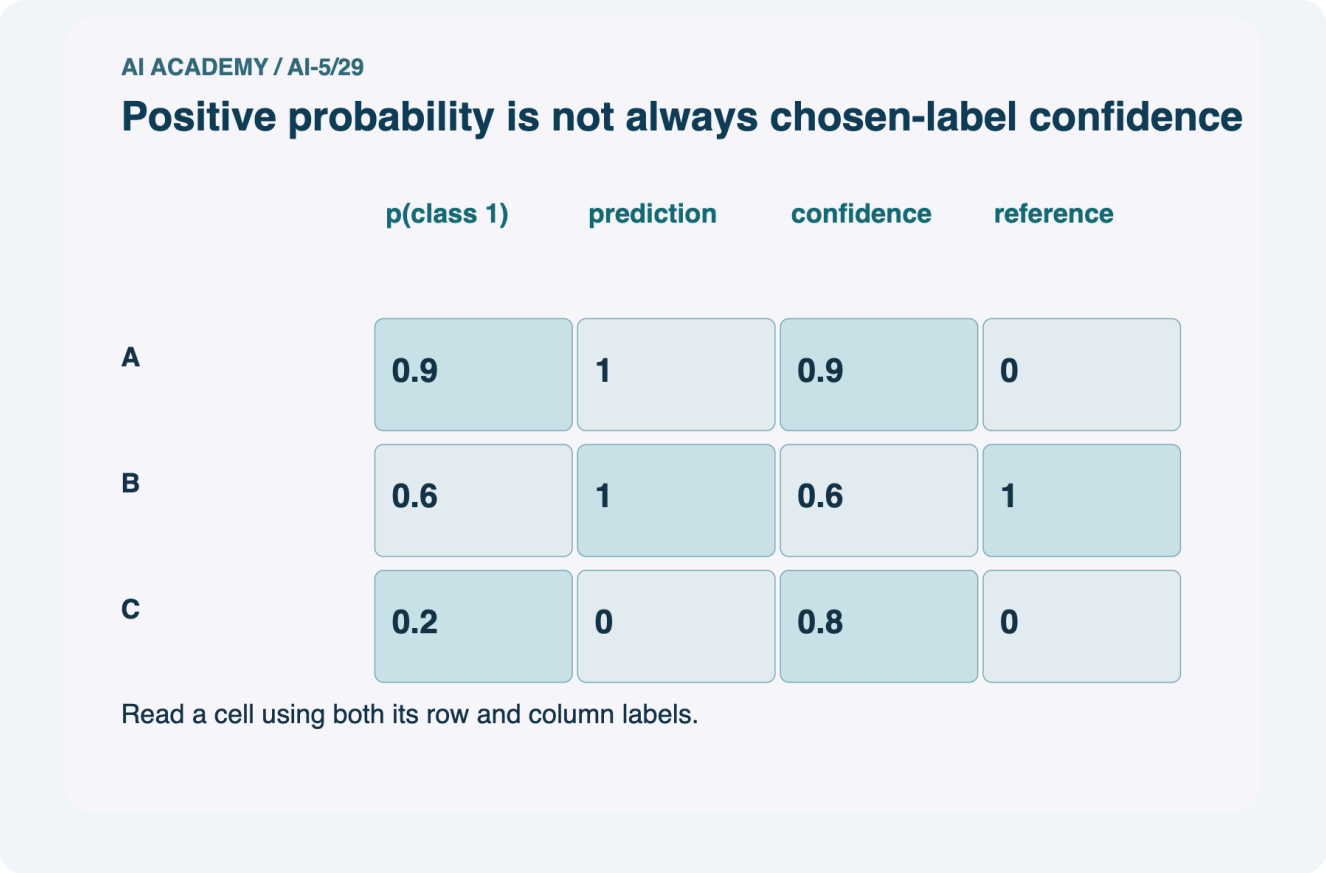
Predictions: A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct. → Selected-label confidence: Confidences are 0.9, 0.6, 0.8.

C's confidence is 0.8 despite p(class 1)=0.2. → Selective review: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Positive probability is not always chosen-label confidence



Follow the labelled picture with the matching explanation. Open the vector diagram

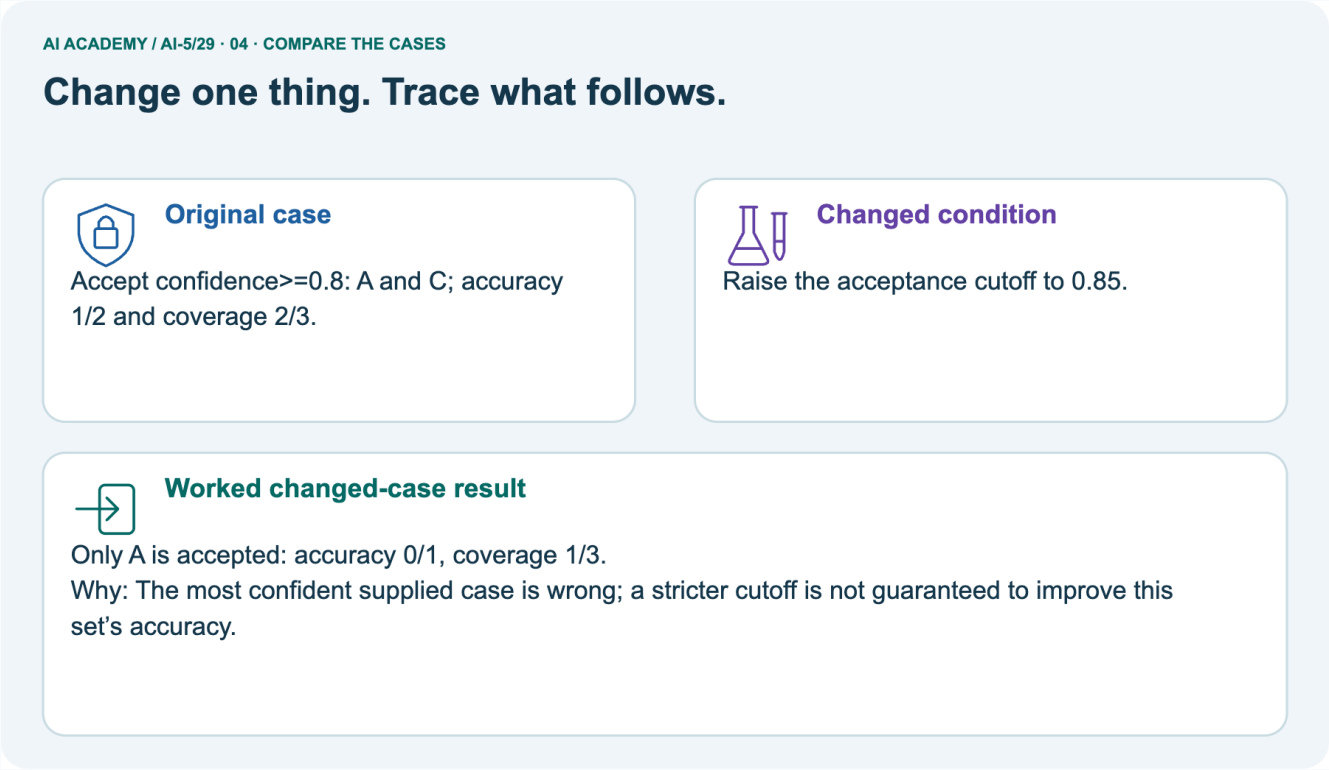
Positive probability is not always chosen-label confidence. A: p(class 1) = 0.9, **prediction** = 1, confidence = 0.9, reference = 0. B: p(class 1) = 0.6, **prediction** = 1, confidence = 0.6, reference = 1.

C: p(class 1) = 0.2, **prediction** = 0, confidence = 0.8, reference = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

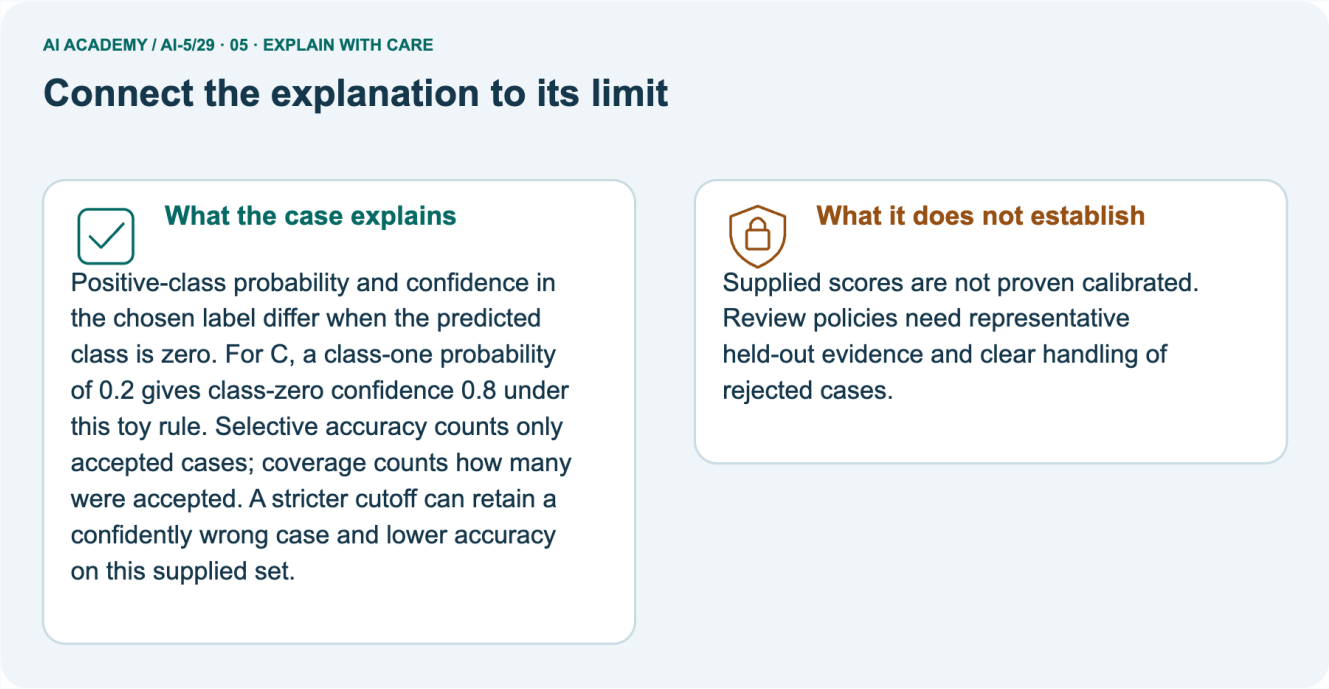
Original case: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3. Changed condition: Raise the acceptance cutoff to 0.85. Expected result: Only A is accepted: accuracy 0/1, **coverage** 1/3.

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

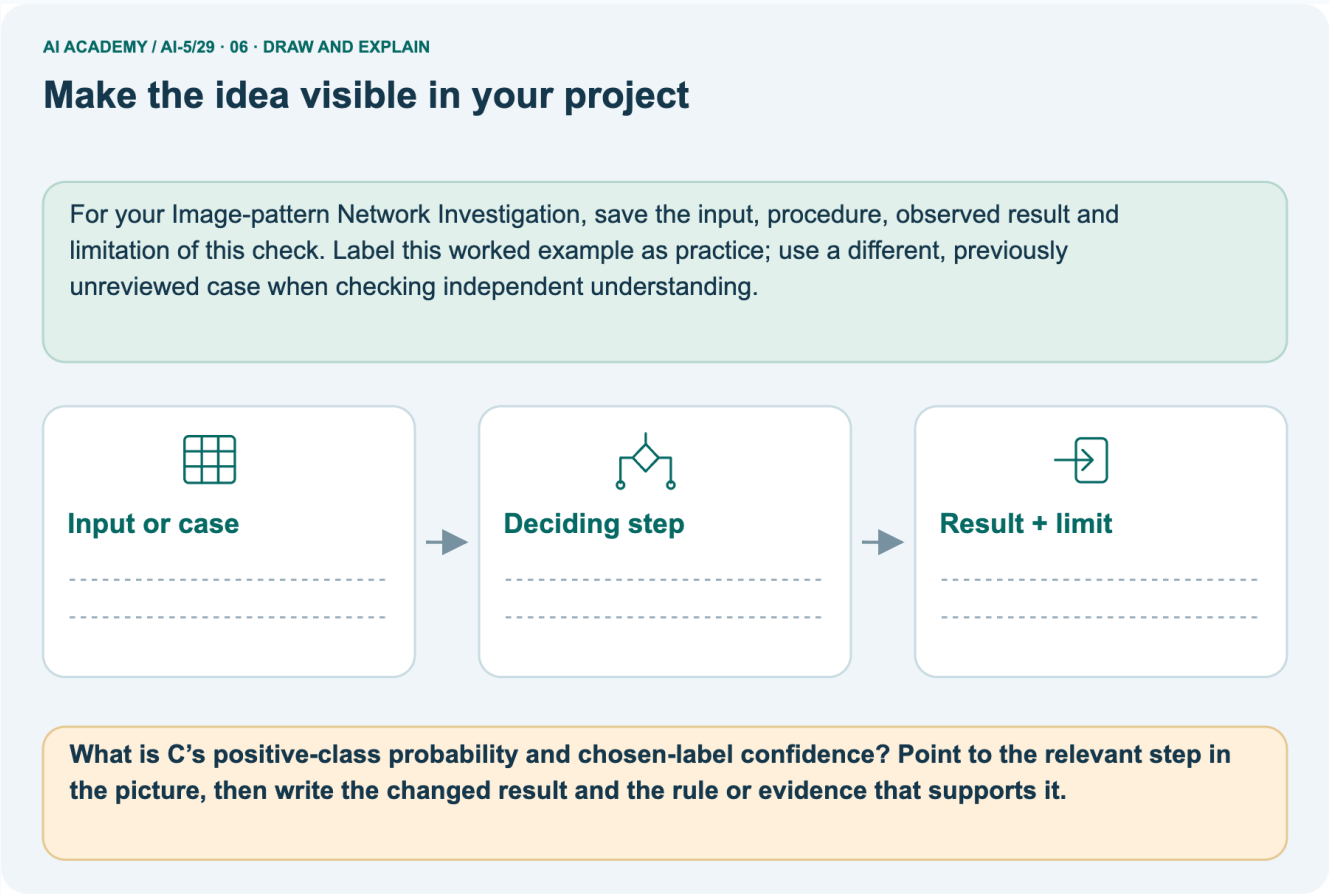
Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Limit: Supplied scores are not proven calibrated. Review policies need representative held-out **evidence** and clear handling of rejected cases.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is C’s positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.